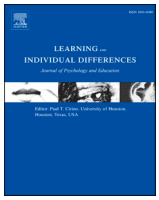
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Skill learning as a predictor of literacy abilities and related impairments among second-graders from low-SES

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ABSTRACT

This longitudinal study tested a conceptual model involving children's motor-skill learning, cognitive measures, and word-level literacy measures as predictors of higher levels of literacy skills. Motor skill learning was assessed with the Invented Letter Task (ILT), representing procedural learning. Second graders (n = 124) from low-SES schools practiced skill-learning across two consecutive days and two weeks later. Midway through the year, fine-motor coordination, working memory, phonological awareness, vocabulary, spelling, writing, and reading fluency were assessed. National standardized literacy tests were administered at the end of the school year. Using structural equation modeling, skill-learning first-day achievements (signifying learning) and two-week post-training performance (signifying automaticity), along with measures assessed midyear, served as predictors of reading comprehension and written expression, confirming the contribution of skill-learning to literacy acquisition. The risk for impairments in reading comprehension/written expression was tested using the same model, indicating similar, but not identical, associations.

Educational relevance statement: This study underscores the importance of skill-learning abilities for literacy development in second-grade children from low-income families. By using a letter-writing task, end-of-year reading and writing performance were predicted. The research identified key factors that may indicate which children are at risk for literacy impairments, emphasizing the need for early interventions. Early training in basic skills is shown to have significant long-term educational benefits. These findings suggest that educators can use such tasks to screen and support students, thereby improving literacy outcomes.

1. Introduction

The acquisition of reading and writing skills requires extensive practice, which involves the procedural learning system (PLS) to transform these skills into procedural knowledge. The knowledge is then stored in the long-term procedural memory system (Dorfberger et al., 2007; Ferbinteanu, 2019; Julius & Adi-Japha, 2015; Kaufman et al., 2010; Kóbor et al., 2018; Nicolson & Fawcett, 2007; Ullman, 2004, Ullman et al., 2020). The PLS, a domain-general learning mechanism, is activated when repeated learning experiences are encountered in motor, perceptual (Censor et al., 2012), cognitive (Fiser & Aslin, 2002), and linguistic domains (Iverson, 2010; Ullman & Pierpont, 2005; Vakil et al., 2015; Vicari et al., 2005). Impaired PLS functioning may help explain the dysfunction of children with developmental learning disorders (Nicolson & Fawcett, 2007, 2011, 2019). The present study looked at monolingual Hebrew-speaking school-aged children with a wide range of reading and writing abilities from low socioeconomic status (SES) families, across their second grade. The study investigated the association between domain-general skill-acquisition learning mechanisms and domain-specific literacy skills to predict reading comprehension and written expression, as well as the risk for impairment in attaining these abilities.

Research stresses the role of poverty in widening academic achievement gaps (Duncan & Murnane, 2011), requiring supplemental studies to predict and facilitate support in the acquisition of reading and writing for those living in poverty (Niedo, Lee, et al., 2014). While

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linguistic, cognitive, and motor abilities required for literacy acquisition have been thoroughly studied in reading and writing impairments, including in Hebrew-speaking children, the role of skill learning in conjunction with these variables has not been studied yet, especially not in children from low-SES families.

1.1. Linguistic, cognitive, and motor factors affecting literacy acquisition

Literacy acquisition is shaped by multiple underlying components. Some of them, as phonology, morpho-syntax, and fine motor coordination, require consistent practice. Phonology enhances sound recognition, morpho-syntax governs word structures and sentence formation, ensuring grammatical accuracy, and fine motor coordination supports writing skills. Together, these abilities form the foundation for successful reading and writing.

Studies of reading impairments in Hebrew initially focused on the phonological deficit of readers (e.g., Bentin, 1992). More recently, however, emphasis has been given to the morpho-syntax level, as Hebrew is a highly inflected synthetic language with an abundance of inflectional morphology (Shechter & Share, 2025). Morpho-syntax deficits of readers are not explained by phonological deficits or working memory (Schiff et al., 2016; Shalev-Leifer, 2016) and appear not only in adults but also in elementary school children (Schiff & Ravid, 2004, 2007; for a recent review of learning to read and developmental dyslexia in Hebrew see Shechter & Share, 2025). Low SES was found to additionally impede these language skills (Levie et al., 2017).

Studies of disorders in Hebrew writing focus on (a) handwriting kinematics (e.g., writing rate), (b) letter readability and handwriting legibility; (c) the analysis and categorization of spelling mistakes; and (d) language use (e.g., Friedman & Gvion, 2023; Hen-Herbst & Rosenblum, 2019). These studies further stress the role of fine motor coordination, known to be lower in children coming from low SES (Ratzon et al., 2007), and emphasize the role of practice for achieving mastery.

1.2. 'Learning' vs. 'automaticity' phases in skill acquisition

The study of skill acquisition differentiates between the overall level of performance and performance changes across learning phases. Skill acquisition initially develops relatively quickly across different experimental paradigms; rapid improvement is measured throughout a single training session. The learning rate slows with incremental gains over multiple practice sessions. When performance reaches nearly asymptotic levels, it becomes an *automatic* response. Progression from fast improvement to slow motor learning is thought to depend on appropriate consolidation processes, defined as the progressive stabilization of a recently acquired memory (Dudai, 2004).

Memory consolidation processes are triggered by the training experience but require time to reach successful completion (see review in Censor et al., 2012; Newell et al., 2009). Phases of learning in simple tasks have been extensively studied in adults and adolescents and, more recently, in children (Adi-Japha et al., 2019; Dorfberger et al., 2007; Julius & Adi-Japha, 2015; Savion-Lemieux et al., 2009). To define a skill as 'learned' or 'memorized,' it must be stable at its automaticity levels, after being retained for days or weeks (Hallgato et al., 2013; Janacsek & Nemeth, 2012; Krakauer & Shadmehr, 2006; Nemeth et al., 2010; Robertson, 2009; Song, 2009). In the current study, this long-term learning outcome is termed the 'automaticity phase'.

Failure to retain a skillful level of performance after training on a novel skill has been reported for different developmental disorders, including developmental language disorder (DLD, formerly known as specific language impairment, SLI; Adi-Japha et al., 2011; Desmottes et al., 2016; Hsu & Bishop, 2014; Lum et al., 2014; Ullman, 2016), dyslexia (Folia et al., 2008; Gabay et al., 2012; Lum et al., 2013; Nicolson et al., 2001; Nicolson & Fawcett, 1990, 2011) and dysgraphia (Adi-Japha et al., 2007; Ben-Pazi et al., 2007; He & Tong, 2017; Parush et al., 2010; Rosenblum, 2018). Children who fail to retain skillful performance on the skills that underlie reading and writing could be at risk for impairments in literacy.

1.3. Theoretical frameworks: literacy acquisition and impairments

The current research was informed by three conceptual frameworks that propose different but complementary explanations for reading and writing acquisition and/or impairments: (a) Ullman's (2004) and Ullman et al. (2020) *procedural deficit hypothesis* focuses on the association between the learning of novel skills using the PLS and language-related abilities; (b) Nicolson and Fawcett's (2007, 2011, 2019) *procedural learning deficit hypothesis* focuses on the automaticity phase of skill acquisition and their *delayed neural commitment hypothesis* suggests that a variety of cognitive networks may be implicated in skill learning deficits; and (c) Berninger's (2009; Berninger, Richards, & Abbott, 2015) model of levels of literacy. These frameworks will be described with respect to how they inform the conceptual model put forward in the current study.

1.3.1. Initial skill learning: the procedural deficit hypothesis

Ullman (2004) and Ullman and Pierpont (2005) formulated the Procedural Deficit Hypothesis (PDH). The PDH suggests that the PLS is associated with the acquisition of some linguistic skills (e.g., grammar and speech-sound representations). Therefore, a procedural deficit may account, at least partially, for impairments in grammar and motor-speech disorders, and consequently in DLD and dyslexia, due to the involvement of the PLS in the acquisition of these specific linguistic skills. For example, Jiménez-Fernández et al. (2011) has shown that children with dyslexia show deficits in implicit sequence learning by the end of a training session, thereby suggesting procedural learning is impaired. Procedural learning deficits may be implicated in early learning, consolidation, or automatization, depending on the brain region and type or severity of impairment. The PDH emphasizes the role of initial learning. For example, in dyslexia, the early phase learning of reading-related abilities as the learning of perceptual sequence, and category learning were shown to be impaired. These abilities may affect the learning of speech-sound representation through rapid processing and result in a lack of cohesion in phonemic category representations (Ullman et al., 2020), affecting spelling (Earle & Ullman, 2021). Furthermore, atypical morpho-syntactic development and processing found in dyslexia may be attributed to a more general deficit in grammar learning and processing during initial grammar learning.

1.3.2. Automaticity: the procedural learning deficit hypothesis

Nicolson and Fawcett (1990) initially proposed that dyslexia may be linked to deficits in the automaticity phase (vs. initial learning), which was refined in the cerebellum deficit hypothesis (Nicolson et al., 2001). Nicolson and Fawcett (2007, 2011) extended and generalized these findings to other developmental disorders (e.g., dysgraphia), reformulating previous frameworks to produce *the procedural learning deficit hypothesis*, which proposes that individuals with these developmental disorders may suffer from an inability to automatize new procedural skills. For example, individuals with dyslexia find it hard to automatize phonological skills. This hypothesis has been investigated by extending skill-learning studies beyond the initial day of training. In individuals with Dyslexia, consolidation which is typically assessed 24 h after the initial training experience, may be delayed or impaired (Gabay et al., 2012; Hedenius et al., 2013; Nicolson et al., 2010; Vicari et al., 2005). Thus, even after extensive training, the *quality* of automatic performance, as assessed by speed and accuracy of performance, may be impaired in individuals with dyslexia (Nicolson & Fawcett, 2019).

1.3.3. Delayed skill acquisition: 'the delayed neural commitment' hypothesis

Nicolson and Fawcett (2019) introduced the 'Delayed Neural Commitment Framework' (DNC), suggesting that developmental

learning disorders may reflect slower skill learning due to delays of the neural network's 'commitment' caused by increased neural noise at earlier stages of network synchronization. The DNC is conceived as a 'forward model,' suggesting that interference in the early developmental stages of the brain's neural network construction results not only in generally slower skill learning but also in longer building (and rebuilding) of the neural networks that underpin reading acquisition. Examples include a delay in the acquisition of early developing language-related skills (e.g., phonology); difficulty in moving from word-decoding to whole-word reading; and difficulty with executive functions needed for reading. In the current study, cognitive-skill 'representatives' of different networks involved in literacy acquisition (i.e., phonology, vocabulary, reading fluency, working memory, and fine motor skills) are included in our conceptual model.

1.3.4. Specific learning disorders: 'cascading levels of language'

Deriving from intervention studies, Berninger, Richards, and Abbott (2015) construct a conceptual framework that differentiates the three most common specific learning disorders in school-age children in relation to three levels of linguistic processes—*dysgraphia* (impaired handwriting at the subword level), *dyslexia* (impaired word spelling and reading), and *oral and written language learning disability* (impaired oral and written syntax comprehension and expression). The framework is based on studies involving relations between different levels (i.e., subword, word, and grammar/syntax/text) of the reading and writing systems, whereby the acquisition of reading skills (for example, reading comprehension, representing "language by eye") and writing skills (for example, written expression, representing "language by hand") is supported by two additional language systems: aural comprehension ("language by ear") and verbal expression ("language by mouth"). Reading comprehension and written expression are considered higher-level literacy skills, with bidirectional relations at the level of word recognition and transcription skills (i.e., spelling and letter writing automaticity), facilitating their concomitant acquisition. The acquisition of these skills is further supported by phonology and grammar/syntax skills at the text level (Berninger & Abbott, 2010; Berninger et al., 2002) and related working-memory skills (Berninger, Swanson, & Griffin, 2015).

1.3.5. Summary: theoretical integration

Based on the framework of Berninger et al. (2002, 2010) and Berninger, Swanson, and Griffin (2015) and the models put forward by Ullman et al. (2020) and Nicolson and Fawcett (2019), the model in the current study examines two different levels of the PLS: initial learning and automaticity, assessed using a novel letter-level task along with working memory to predict later achievements in reading comprehension and written expression. The model addressed three levels of linguistic processes: subword (letter), word, and text level.

Our main goal is to examine whether skill acquisition processes, specifically learning and automaticity, are associated with the underlying processes of literacy acquisition that support reading comprehension and written expression. Moreover, we investigate whether these underlying processes can predict the risk for impairment in reading comprehension and written expression.

1.4. The Invented Letter Task (ILT)

To represent the "learning" and "automaticity" phases of procedural learning, we use the Invented Letter Task (ILT; Julius & Adi-Japha, 2015). The ILT is a grapho-motor skill-learning task in which children repeatedly practice the writing of an "invented" letter across horizontal lines. Handwriting is considered a procedural motor skill (Dayan & Cohen, 2011). ILT practice extends over two training days, and retention is assessed for 2 weeks post-training. Here we refer to two measurements of progression in ILT acquisition: end-of-training on Day 1 (here termed "learning" phase) and 2 weeks post-training (here termed "automaticity" phase).

The ILT 'learning' phase corresponds to the early stages of procedural knowledge acquisition. It may be associated with underlying writing and reading processes supported by the PLS, such as learning correct speech-sound representation and phonemic cohesion, related to spelling abilities (Ullman et al., 2020).

The association between ILT automaticity measures and literacy may depend on the specific role of letter automaticity in contributing to literacy skills, including written expression (Berninger et al., 2002). The rate and accuracy of skilled letter writing across rows in the 'automaticity' phase may be associated with fluent writing. Moreover, ILT 'automaticity' represents PLS automaticity. Therefore, the ILT automaticity phase may be further associated with reading comprehension and written expression, abilities supported by the automaticity of reading processes at the subword and word levels (Ullman et al., 2020).

Hebrew writing involves disconnected letters. Like other Semitic languages, Hebrew is written and read from right to left. The ILT, like reading and writing in Hebrew, requires parsing discrete elements and horizontal progression between elements. Recent studies showed a direct link between ILT learning and reading and writing skills in Hebrew and Arabic (Ghanamah & Adi-Japha, 2023, Ghanamah et al., 2024; Julius et al., 2016; Salameh-Matar et al., 2024). For example, Julius et al. (2016), who tested 56 Hebrew-speaking children from kindergarten and second grade from families of medium to high SES, found associations between task performance and concurrent as well as following-year reading fluency, and handwriting fluency and legibility. The strength of the association depended on the task (writing/reading) and ILT phase. The association between the ILT and literacy measures aligns with other studies that show associations between letter copying tasks and literacy measures (e.g., Berninger et al., 2002; Suggate et al., 2018).

1.5. The current study

Literacy acquisition is a multifaceted process shaped by various interconnected elements, including procedural, declarative, and working memory, socio-emotional influences, environmental factors, and language-specific characteristics. The current study focuses on the role of procedural memory in supporting literacy-related linguistic, cognitive, and motor skills and reflects the letter, word, and text levels.

The study follows children from the beginning to the end of second grade, encompasses a wide array of explanatory factors, and focuses on literacy skills to explain reading comprehension and written expression assessed at the end of second grade, as well as the risk for attaining low achievements in these skills. The study sample involved children from a low-SES population. Throughout second grade, children in the lowest 30 percentiles for reading scores may attain average scores (Spira et al., 2005). This suggests that labeling a child as having a reading or writing disorder may be premature, and children with lower abilities should more accurately be described as "at risk".

We hypothesized that the ILT learning and automaticity phases (assessed at the beginning of second grade) would be associated with reading comprehension and written expression (assessed at the end of second grade), along with basic the linguistic, cognitive and motor literacy-related abilities (assessed at the middle of second grade) as possible mediators of the association between the ILT and reading comprehension and written expression scores (Fig. 1). Because children learn to read and write in the early grades of elementary school, we focus on concomitant prediction of these outcomes. Additionally, a risk for impairment in reading comprehension or written expression was separately studied as an outcome of the model. It is worth noting that factors predicting an ability and those predicting the risk for impaired ability may not always coincide.

Current research practice recommends multi-level statistical analysis with overlapping domains rather than separation of learning domains (Archibald et al., 2019; Peters & Ansari, 2019). Therefore, structural

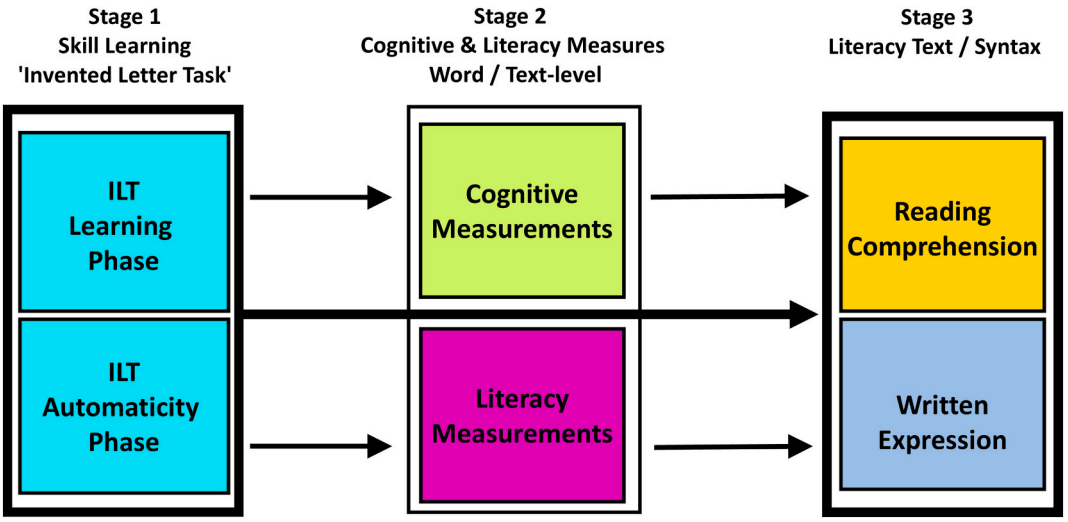


Fig. 1. Procedural learning prediction of literacy abilities, theoretical model.

equational modeling (SEM) was used.

The model included the following measures that were found in previous studies to be significant predictors of literacy attainments in both reading and writing: A) motor learning processes ILT (Julius & Adi-Japha, 2015; Julius et al., 2016); B) fine motor coordination (Beery's subtest of motor coordination) (e.g., Adi-Japha and Abu-Asba, 2014; Costa et al., 2018; Dinehart & Manfra, 2013; Grissmer et al., 2010), C) cognitive and verbal measures: (1) working memory (Berninger et al., 2010; Schuchardt et al., 2008) (2) phonemic awareness (Bradley & Bryant, 1983; Caravolas et al., 2019; Share et al., 1984; Snowling, 2001; Spira et al., 2005; Torgesen & Mathes, 1999; Wagner et al., 1997), (3) vocabulary (WISC R-95) (Berninger, Richards, & Abbott, 2015; Greenblatt et al., 1990); D) literacy measures including: (1) text-reading fluency (Fuchs et al., 2001; Kim et al., 2010; Schoonen, 2019; Shany & Share, 2011, Share, 2008; Valencia et al., 2010), (2) spelling (Niedo, Abbott, & Berninger, 2014; Ravid, 2012), and (3) text writing (copying) speed (Graham, 1986; Erez & Parush, 1999; Nicolson & Fawcett, 2011, Nicolson & Fawcett, 2008; Parush et al., 2010; Rosenblum, 2008; Volman et al., 2006). Predicted literacy skills include reading comprehension and written expression (Berninger, Richards, & Abbott, 2015; Berninger, Swanson, & Griffin, 2015; Caravolas et al., 2019; Niedo, Lee, et al., 2014; Primor et al., 2011).

2. Method

2.1. Participants

A sample of 151 second-graders from two elementary schools from the same city in Israel took part in the study over three consecutive years. The schools' socioeconomic status (SES) was classified at national ranks 8 and 9 (on a 1–10 scale, with 10 as the lowest), i.e., low-SES. The children in these schools took part in a national nutrition program that entitled them to subsidized lunch. All second-graders were invited to participate. The inclusion criteria were: monolingual Hebrew speakers; nonverbal reasoning above 85 (Raven et al., 2008; see Section 2.2.4); and no pharmacological treatment during the years of the study. The final sample included 86 boys and 38 girls (N=124, M_{age}=7.93; SD=0.50). Parents of the child participants completed a questionnaire (see Section 2.2.1). See Table 1 for a socio-demographic description of the sample.

The study was conducted in 2016–2018 and received ethical approval from the Ministry of Education (7883/054 10.32). Schools, parents, and children were informed about the study. Parents have signed informed consent forms for their participation and their child's participation.

Table 1
Descriptive statistics of the sociodemographic data at pretesting.

Measures	Mean	SD	Range
Age (years)	7.93	0.50	7.01–9.46
Education father (years)	14.02	3.48	8–25
Education mother (years)	13.15	1.82	8–20
Profession father (ranks 1–5)	2.63	1.59	1–5
Profession mother (ranks 1–5)	2.50	1.51	1–5
Number of children in a family	6.31	2.57	1–14
Number of rooms	4.24	1.14	1–8
Income (ranks 1–5)	2.01	0.97	1–5
Nonverbal reasoning[a]	105.21	10.38	87–129

Note. Age (years)—according to Stage 1. Standard scores are expressed for: Nonverbal reasoning—Raven's Educational Colored Progressive Matrices (CPM); Fine motor coordination—number of correct items (maximum 30, in 5 min) on the Visual Motor Integration (Beery) Motor subtest; Vocabulary—number of correctly described words on the WISC-R 95 subtest.

[a] Nonverbal reasoning = Raven IQ scores assessed at the beginning of stage 2.

2.2. Measures

2.2.1. Parent demographic questionnaire

Before the start of the second grade, parents completed a questionnaire describing their children's development and family background. To code socioeconomic status, a seven-factor index based on Hollingshead (1975) was adapted to Israeli culture by Korat et al. (2007). One parent of each child participant reported education levels, occupations, and/or professional qualifications for each parent, number of children, number of rooms, and household income. The data were transformed to a 5-point scale (from 1=*low* to 5=*high*). The professional qualification and current-occupation scale ranged from 1 (*unskilled workers and menial laborers*) to 5 (*upper-level executives and professionals*). Parents ranked their combined income relative to the official Israeli average at the time of testing on a scale from 1 (*far below the national average*) to 5 (*far above the national average*).

2.2.2. Invented Letter Task (ILT)

The ILT task stimuli consist of point-to-point planar movements in one line (Fig. 2A shows one stimulus letter and 2C shows one child-written letter). Writing direction is: A→B→C, segment length 1.2 cm, circle outer diameter 3 mm, shape width 6 mm) (Adi-Japha et al., 2011; Hollander & Adi-Japha, 2021; Julius & Adi-Japha, 2015). Movement

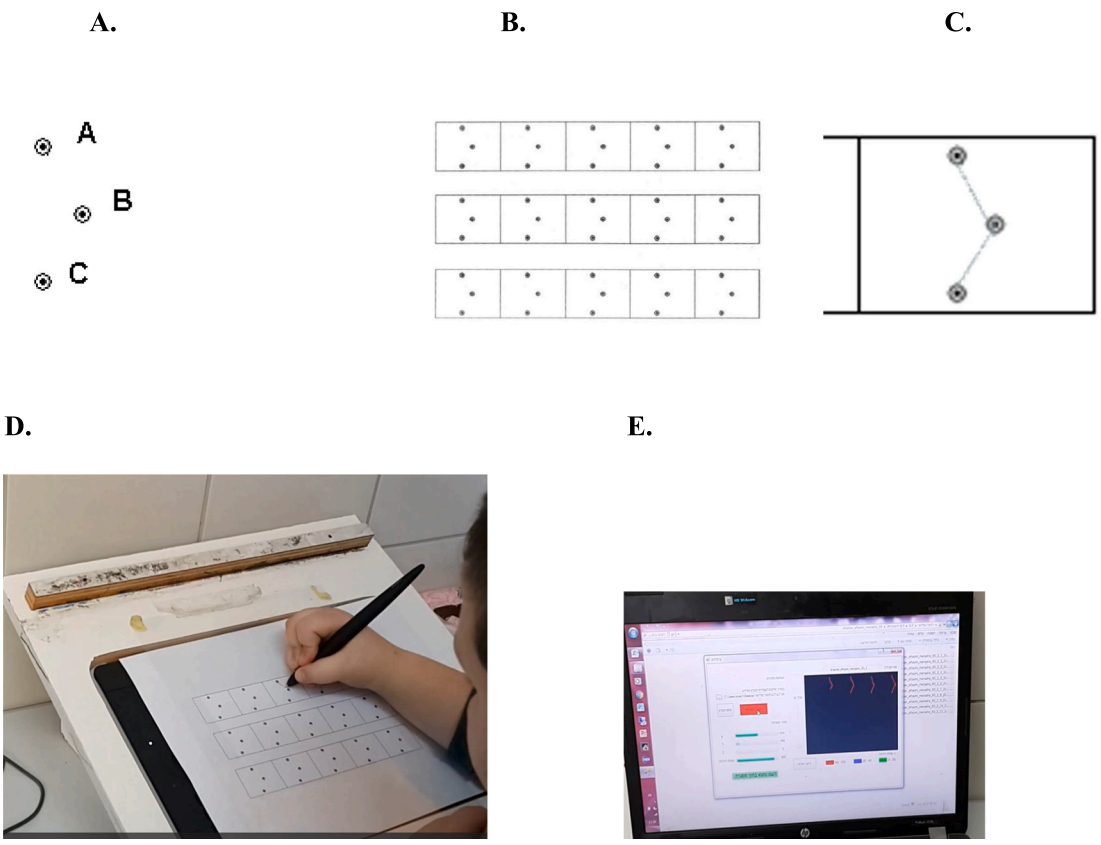


Fig. 2. The Invented Letter Task. Panel A shows an individual ILT Stimulus. Panel B illustrates one ILT Block. Panel C. One written letter. Bottom panel D—A child completing the ILT with the Wacom tablet and an ink stylus pen. The child is placed in front of the ergonomic table, to fit the tablet digitizer for measuring ILT production. The Wacom tablet is placed on the board; an ILT block page is placed and glued to the board and the tablet so the page is affixed while the task is performed. Bottom panel E—the computer screen showing the production online. The MATLAB and Java software, dedicated to the ILT, is computing rate and accuracy.

progresses from right to left (as in Hebrew writing).

Each ILT block contained 15 repeats of the same stimulus (Fig. 2B). The blocks were separated by a minimum of 15 s and up to a 30-second break at all three intervals. The children were asked to complete the task as quickly and as accurately as possible, going through all three encircled dots with one line. General encouragement was provided between blocks ('You're doing fine'; 'Remember to be as quick and accurate as possible!'), but no feedback on performance was given.

The ILT was used to study the time-dependent course of skill acquisition across two consecutive days. Retention was assessed 2 weeks later (Fig. 3). The task was performed in 28 blocks: 12 blocks in the first-day training activity (the last 4 blocks are referred to as the 'learning phase'), 12 blocks after 24 h (i.e., the second-day training activity), and four blocks two weeks after the first-day training activity to study skill retention (referred to as the 'automaticity phase'). The children practiced the task at the beginning of each day for one block until they completed one line with no errors. The children completed all ILT blocks within 20 min.

2.2.2.1. Apparatus. The ILT block was printed on an A4 sheet of paper and placed on a Wacom tablet (INTUOS Pro small [PTH-451], 320 × 208 × 12 mm., 50 degrees of tilt sensitivity). Transparent chrome paper covered and attached to the tablet, revealing the exact active surface area (157 × 98 mm) for production of the block. The cordless tablet was placed on a slanted tailored ergonomic board adjusted to the child's seat height to achieve the recommended ergonomic writing position (a 20-degree slant). The children rested their feet on a stable platform (in a 90-degree position). An ink stylus pen that left a visible ink trace on the

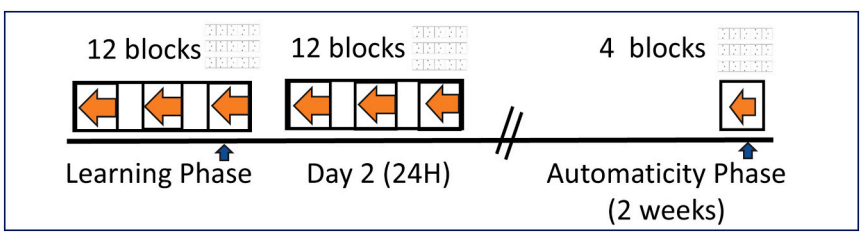


Fig. 3. The Invented Letter Task design. The timeline of the ILT is shown in this figure divided into 3 days. The orange arrows in each time point, describe the Movement progresses from right to left in each block (as in Hebrew writing). The ILT consists of 12 blocks of Fig. 2B on Day 1 and 2 and an additional 4 blocks two weeks later. The term 'Learning Phase' refers to the last 4 blocks in Day 1, marked with the small blue arrow. The term 'Automaticity Phase' refers to the 4 blocks scored two weeks post-training, marked with the second small blue arrow. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

printed block was used. The children saw their product on the page; the experimenter also saw it on the computer screen and was able to monitor the production process (see Fig. 2D and E). After each ILT block was completed, the experimenter placed an identical sheet of paper on the tablet in front of the children, asked the children if they were ready, and gave them a sign to start.

2.2.2.2. ILT coding. The tablet forwarded information to the computer at 200 Hz. MATLAB and Java software dedicated to the specific ILT computed rate and accuracy (see Fig. 2E). Rate (writing duration) was computed in milliseconds from the first touch of the tip of the pen on the page until completion. Accuracy scores were computed at a nominal accuracy of 0.02 mm. Erroneous shapes included those *not* produced in one continuous movement (e.g., a shape composed of two segments) or too narrow or too wide with respect to the mid-point (the mid-point was located 0.6 mm. to the right of the upper inside point on the *X*-axis; shapes were coded as errors if they were ≤0.325 mm or ≥0.875 mm wide). The number of errors in each block was coded as the block's accuracy score. Rate and accuracy were calculated at two time-points: (a) the 'learning phase' (the average of the last four blocks on Day 1) and (b) the 'automaticity phase' (the average of the four blocks two weeks after the learning phase). A significant difference was found for Rate between the learning automaticity phases ($t(122)=5.62$, $p<.001$). Children improved in the two weeks between the learning phase (Mdn=30.61 s.) and the automaticity phase (Mdn=27.35 s.) while maintaining the number of errors ($t(122)=-0.67$, $p=.50$) (Mdn=1.25, Mdn=1, respectively), replicating the results of previous studies using the ILT (Julius & Adi-Japha, 2015). Rate and Accuracy were converted to Z scores. The total quality score used in the model is the averaged Z-score of the rate and accuracy measures at each time-point.

2.2.3. Fine motor coordination

The Beery-Buktenica developmental test of Visual Motor Integration (Beery & Beery, 2010, 6th ed) motor-coordination (MC) subtest was chosen due to its strong correlation with writing skills, its extensive use in research and practice, and its similarity to the ILT. The MC subtest is standardized for individuals between ages 2 and adulthood, and it includes 30 geometric forms to be traced in a trail of connecting dots. Forms are set in a progressively difficult sequence. The test was stopped after 5 min or if the child failed to correctly copy three consecutive shapes. The final score was the number of shapes copied correctly (i.e., raw scores were used). The inter-scorer reliability was found to be 0.94, for 7–9 year-olds Cronbach's $\alpha = 0.84$, and test-retest coefficient was.85 (Beery & Beery, 2010, 6th ed).

2.2.4. Nonverbal reasoning

Raven's Educational Colored Progressive Matrices (CPM) was used as a standardized measure of nonverbal reasoning ($M=100$, $SD=15$), suitable for children aged 5–11 (Raven et al., 2008). Children with scores <85 were excluded from the final analyses. This cutoff was chosen to exclude scores of children at risk for intellectual disabilities (van Wingerden et al., 2018).

2.2.5. Phonological awareness

The Maakav test (Shany et al., 2003) is a norm-referenced literacy evaluation for grades 2–6, recommended as part of the elementary school curriculum in Israel. The phonological-awareness subtest (phoneme deletion) measures children's ability to segment 16 spoken real words into phonemes. Each word is said separately (e.g., 'Say *xatul* [cat]'; 'Now say it without the *x* '. Response: *atul/*). None of the outcomes of this phoneme-level word manipulation was a real Hebrew word. Raw scores (number correct) were used in the analyses (Cronbach's α=0.87).

2.2.6. Working memory

The second part of Subtest 11 of the WISC R-95 Hebrew version of the Wechsler Intelligence Scale for Children (Wechsler, 1998) is a backward digit span test that measures working memory and predicts typical reading and writing achievement (Niedo, Abbott, and Berninger, 2014). Participants were asked to repeat two series of two to eight digits in reverse. Each trial was coded as recalled correctly or incorrectly, and the total number of correct digits across all trials was recorded. Raw scores were used in the analyses (Cronbach's α=0.85).

2.2.7. Vocabulary

Vocabulary was measured using Subtest 7 from the WISC R-95 Hebrew version of the Wechsler Intelligence Scale for Children (Wechsler, 1998). In this progressively difficult test, participants explain the meaning of 25 words read aloud. This measure is one of the best predictors of academic achievement (Berninger, Swanson, & Griffin, 2015; Greenblatt et al., 1990). The maximum score is 50 points. Raw scores were used for the analyses (Cronbach's α=0.85).

2.2.8. Text reading fluency

The text reading fluency subtest of the Maakav test (Shany et al., 2003) involves the presentation of a short narrative story (194 words) designed for second- and third-graders. Children were asked to read it aloud as quickly as possible. Fluency (number of words correctly read in one minute) is a strong predictor of reading comprehension (Kim et al., 2010). Convergent validity ($r=0.59$). Raw scores were used.

2.2.9. Spelling

Spelling was assessed using the dictation-to-writing subtest of the Maakav test (Shany et al., 2003). A text comprising 27 words was dictated, and participants wrote the words. The number of spelling errors was recorded. Raw scores were used (Cronbach's α=0.59).

2.2.10. Text writing rate (copying task)

Text-writing rate was assessed using the Hebrew Handwriting Evaluation (Erez et al., 1996). This standardized assessment allows speed efficiency, and quality of handwriting to be evaluated. Children copied a short story (30 words). The story included all the letters of the alphabet. Raw scores (seconds) of the child's average rate of copying per word of the text were computed and used in the final analyses (Cronbach's α=0.81).

2.2.11. Reading comprehension and written expression

Reading comprehension and written expression were assessed using subtests of the National Academic Reading and Writing ('Meitzav') test for Grade 2—Part A, 2012 adopted to the school culture. Meitzav is part of a national testing program conducted by the National Authority for Measurement and Evaluation in Education, in which elementary, junior high, and high schools participate. Testing in second grade takes place in the last trimester of the school year. The Meitzav test is comprised of six sections (dictation accuracy, reading, and writing accuracy, reading comprehension, written expression, and linguistic knowledge).

The two subsets were scored by a well-experienced literacy teacher and research assistants. The intraclass correlation coefficient (ICC) for inter-rater reliability was 0.80, indicating strong agreement among raters that meets the acceptable threshold for research purposes ($p < .05$). The reading comprehension subtest has a standardized national calculation considering the school's SES level.

2.2.11.1. Reading comprehension. In the reading comprehension subset participants read a story for their age level and answered related questions. Scoring followed the national detailed measures that included the following four items: (a) sentence completion (measures ability to understand sentence patterns, predict appropriate words, and demonstrate syntactic awareness); (b) explicit meaning/literal comprehension

(measures ability to identify and understand information directly presented in text, including main ideas, details, sequences, and explicitly stated relationships); (c) inferential comprehension description (measures ability to draw conclusions, identify underlying themes, understand relationships, and make predictions based on textual evidence); (d) application, evaluation and critique of the text (requires readers to engage with text not just as passive recipients but as active, critical thinkers who can analyze, evaluate, and apply the information they encounter). Raw scores were used in the data analyses. Max. row score was 46.

2.2.11.2. Written expression. The written expression subtest of the Meitzav test includes the presentation of six pictures. The student is requested to write a story according to the pictures. The score includes the evaluation of two elements: content and structural elements, as well as linguistic elements. Content includes quality and depth of ideas presented; relevance to the writing task; supporting details and examples; and completeness of narrative. Structural elements include logical organization of ideas; coherent paragraph development; clear transitions between ideas; appropriate text structure for the genre; introduction, and conclusion effectiveness. Linguistic elements of written expression include grammatical accuracy, stylistic choices, and technical precision, proper syntax and morphology, vocabulary usage, sentence construction, and adherence to language conventions such as punctuation and spelling. Raw scores were used. Max. row score was 25.

2.2.11.3. Risk for impairment in Reading comprehension/written expression. According to the Manual of the Meitzav test, the cutoff score for defining risk for impairment in each sub-test is 1 SD under the relevant national means. A risk score for reading comprehension/written expression was computed as a binary score corresponding to whether the child was at or below the cut-off score.

2.3. Procedure

The study was carried out in three stages (Fig. 4). In the Pretest Stage (before second grade), parental consent was obtained, and parents completed the demographic questionnaire. In Stage 1 (beginning of second grade), the ILT was administered. In Stage 2 (middle of second grade), cognitive and literacy measurements were administered (Sections 2.2.3 to 2.2.10). In Stage 3 (end of second grade), reading comprehension and written expression were assessed. Each session took up to 45 min. Assessments were carried out in a quiet room on the school grounds by an experienced and professionally trained child assessment occupational therapist (the first author). A table listing all outcome variables, including the source and reference of each variable, the assessment timing, the outcome variables, and the scoring range can be found in the Appendix (Table A2).

2.4. Data analysis

The AMOS Structure Equation Modeling (SEM) program, Version 29, was used to create a path analysis with the maximum-likelihood method. SEM was used to test the theoretical model (Fig. 1) and explore significant paths of relations among Stage 1, motor learning processes; Stage 2, cognitive and literacy measurements; and Stage 3, Reading comprehension and written expression. The required sample size was calculated based on several other studies, all investigating students with and without learning difficulties (e.g., Swanson et al., 2013; West et al., 2018). The complexity of the current model was mitigated by using observed scores instead of latent variables to model relationships between the variables. Notably, although larger samples are recommended for path analysis, the sample size ($N = 124$) meets the recommended criterion of a ratio of at least five participants per parameter (11 parameters; Bentler & Chou, 1988). In the SEM analysis, we handled missing data by choosing the option of estimating means and intercepts by Maximum Likelihood Discrepancy.

Model fit was assessed using the following goodness-of-fit indices (see Hu & Bentler, 1999): chi-square (Tabachnik & Fidell, 2007), the

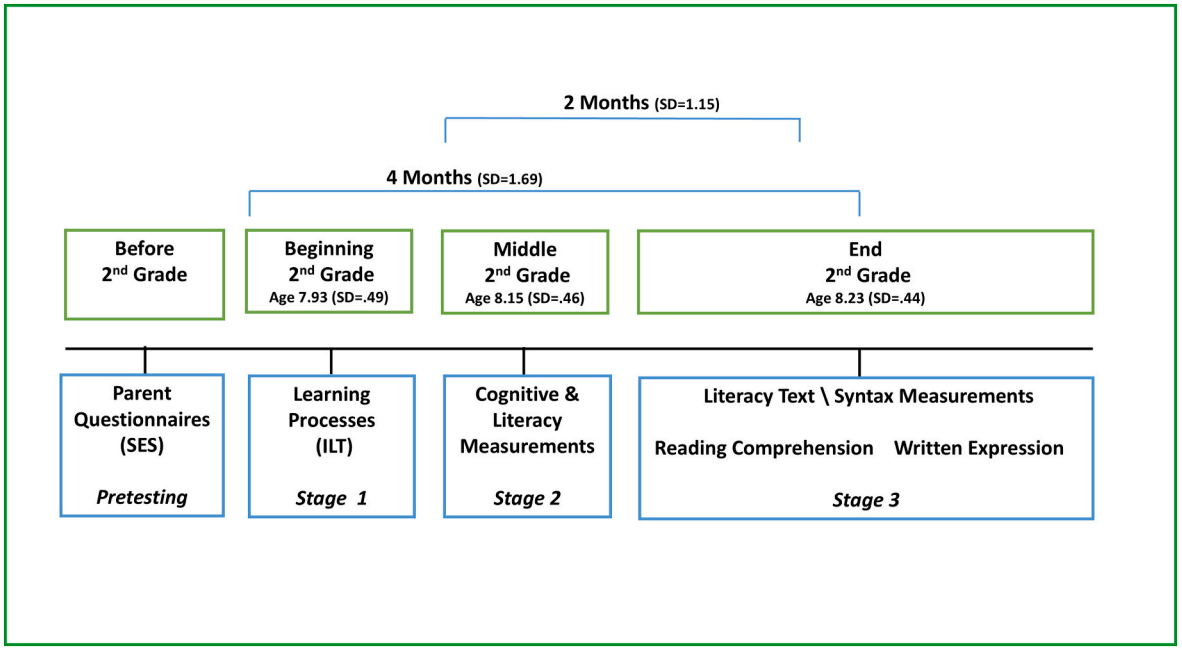


Fig. 4. Data collection timeline and research stages. The stages of the study are presented in the blue boxes, at the bottom of the timeline. The green boxes above the timeline show school-year distribution, children's mean age and standard deviations (*SD*). Gaps between stages are presented in 'months' in blue lines above the boxes with the SD of the children's ages across the stages. Given the small overlap between the stages, a full covariate correlation of age with each stage was calculated. Only phonological awareness showed a small correlation with all stage-age measures (see Appendix Table 1A for correlations with age assessed in Stage 2). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Comparative Fit Index (CFI; Rigdon, 1996), the Tucker-Lewis Index (TLI; Bentler & Bonett, 1980), and Root-Mean-Square Error of Approximation (RMSEA; Browne & Cudeck, 1993). A nonsignificant chi-square, CFI, and TLI, equal to or >0.95, and RMSEA equal to or <0.07 (Hu & Bentler, 1999; Tabachnik & Fidell, 2007) are indicative of an acceptable fit. All variables were within acceptable values for skewness and kurtosis as indicated by the guidelines of Kline (2015).

3. Results

The results are organized into two sections. The first section introduces the structural equation model used to evaluate both mediated and direct paths from the ILT quality scores during the learning and automaticity phases (Stage 1) to reading comprehension and written expression concomitantly (Stage 3). The second section presents the structural equation model used to evaluate both mediated and direct paths from the ILT phases (Stage 1) to risk for impaired reading comprehension/written expression (Stage 3) in two separate models. Descriptive statistics for measures used in the model are reported in Table 2.

Table 2
Means, standard deviations, and ranges for main study measures.

Measures		M (%)	SD	Range
Pretest: *Parent questionnaire*				
SES (Z-score)		0.00	0.55	−1.34–1.53
Stage 1: *InventedLetter Task (ILT)*				
1	Learning phase (Z-score)	−0.00	0.62	−1.30–2.76
2	Automaticity phase (Z-score)	−0.00	0.67	−1.15–3.04
Stage 2: *Cognitive and literacy measures*				
3	Fine motor coordination	19.50	3.20	12–26
4	Phonemic awareness	7.84	3.57	0–15
5	Working memory	3.57	1.33	0–8
6	Vocabulary	12.69	5.23	4–27
7	Text reading (fluency)	71.71	36.73	10.17–161.14
8	Spelling	8.66	5.52	0–21
9	Text writing rate (copying task)	8.21	3.38	3.41–24.83
Stage 3: *Outcome literacy measures*				
10	Reading comprehension	28.10	10.98	2–46
11	Written expression	9.34	6.00	0–21
12	Reading comprehension risk	(28.2 %)		
13	Written expression risk	(25.8 %)		

Note. *N*=124. See Section 2.2 for a description of each measure. Mean (%) of *n*; *SD*—standard deviation; Pre-testing stage: parents' SES questionnaires—Parents Socioeconomic Status questionnaire; Mean Z SES—averaged raw score of all SES measures in the pre-testing stage (Table 1); Stage 1: Motor learning processes learning phase—end of Day 1; Z-score of ILT—(Invented Letter Task)/rate and accuracy measurements were Z-scored and averaged. Automaticity—after two weeks; Stage 2: cognitive and literacy measurements: fine motor coordination—number of correct items on the Visual Motor Integration (Beery) Motor subtest; Sequence verbal WM—Working memory number backward recall of the WISC-R 95 subtest; Phonological awareness—phonological subtest of the Maakav test (number of correct words segmented into phonemes); Vocabulary—number of correctly described words on the WISC-R 95 subtest; Text reading (fluency)—number of correct words read by the child per on text-reading subtest of the Maakav test; Text writing rate (copying task) rate (in s)—calculated for copying a word from the HHE Copy Text subtest; Stage 3: Reading comprehension risk—the reading comprehension subtest of the National Academic Reading and Writing Test (Meitzav Test, 2012); Reading comprehension Risk Cut score (accuracy)—1 *SD* below national mean (percentile <16) among low SES schools (risk row score <23); Written expression the Written Expression subtest of the National Academic Reading and Writing Test (Meitzav Test, 2012), Written Expression Risk Cut score—1 *SD* below national mean (percentile < 16) (risk row score <5).

3.1. Analysis of paths from ILT learning and automaticity phases to reading comprehension and written expression

3.1.1. Correlations

Pearson correlations between precursor and outcome measures were computed. Correlations among Age (calculated separately at the time each measurement was taken), SES score, and all study variables were tested (see Appendix Table 1A); no significant correlations were found in all measures except for a small correlation of age and phonological awareness. These findings rule out the possibility that Age and SES should be used in this study as covariates for predicting reading comprehension and written expression. Therefore, age and SES were not included in the final SEM model.

Table 3 shows the Pearson zero-order correlation among all research variables (v1–13).

It is important to note that outcome scores correlated with most of the study variables. Reading comprehension (v10), correlated with all variables except for the ILT. The risk for impairment in reading comprehension (v12, termed reading comprehension risk in the tables and figures) correlated with all variables except for working memory and the ILT learning phase. Written expression (v11) correlated with all study variables. Risk for impairments in written expression (v13, termed written expression risk in the tables and figures) correlated with all study variables except for fine motor coordination, phonemic awareness, and vocabulary. Thus, reading comprehension and the risk for impairment in reading comprehension had a different set of associations. Similarly written expression and the risk for impairment in written expression had a different set of associations had a different set of associations.

3.1.2. Autoregressive SEM: reading comprehension and written expression

Figs. 5–7 present results for the SEM model that tested transitional relations among the variables. In all measures, only fit indicators and significant standardized path coefficients are presented. Nonsignificant paths and correlations are then removed to produce a simplified model in which all remaining relationships are statistically significant. Estimates of standardized regression weights (beta) are presented under the straight arrows in the model, and correlations are shown on the curved arrows (Pearson's coefficients- r). All statistics are well within the accepted ranges that indicate an excellent fit (Hu & Bentler, 1999).

In Fig. 5, the final data fit measures are: $\chi^2(33)=36.41$, $p=.31$, $N=124$; CFI=0.99; RMSEA=0.03; TLI=0.98; NFI=0.91. Table A3 in the Appendix includes standardized estimates of regression coefficients with their *p* values, the latter also indicated in the figure.

The R^2 values indicate that the model explained 36.0% of the variance of reading-comprehension ($p < .05$) and 49.0% of the variance of written expression ($p < .05$). There was a significant correlation between reading comprehension and written expression ($r = 0.30$). The ILT affects written expression directly (ILT automaticity phase) and indirectly via text reading fluency (ILT learning phase) and spelling (ILT learning and automaticity phases). The ILT affects reading comprehension indirectly via text reading fluency (ILT learning phase) and fine motor coordination and working memory (ILT automaticity phase). It should be noted that the same pattern of associations emerged when reading comprehension only or written expression only was entered as the model outcome (Figs. 1A and 1B in the Appendix).

3.2. Analysis of paths from ILT learning and automaticity phases to risk for impairment in reading comprehension and written expression

Additional models were suggested for the analysis of Stage 1 to Stage 3 direct paths and correlations. These included risk for impairment in reading comprehension as the outcome (Fig. 6) and risk for impairment in written expression as the outcome (Fig. 7). All statistics are well within the accepted ranges that indicate an excellent fit (Hu & Bentler, 1999). The direct paths and correlations showed the same pattern of

Table 3
Pearson zero-order correlation among all research variables.

	1	2	3	4	5	6	7	8	9	10	11	12	13
1 Learning phase (ILT)	1	0.52**	−0.22*	0.16	−0.14	−0.01	−0.20*	0.38**	0.23*	−0.10	−0.31**	0.10	0.26**
2 Automaticity phase (ILT)		1	−0.34**	0.07	−0.19*	0.03	−0.16	0.34**	0.28**	−0.14	−0.33**	0.22*	0.34**
3 Fine motor coordination			1	−0.08	−0.01	0.07	0.12	−0.21*	0.02	0.28**	0.19*	−0.32**	−0.14
4 Phonemic awareness				1	−0.04	−0.14	−0.24**	0.30**	0.15	−0.37**	−0.19*	0.32**	0.13
5 Working memory					1	0.16	0.13	−0.33**	−0.08	0.25**	0.24**	−0.13	−0.21*
6 Vocabulary						1	0.30**	−0.15	0.04	0.36**	0.18*	−0.28**	−0.08
7 Text reading (fluency)							1	−0.59**	−0.39**	0.52**	0.61**	−0.46**	−0.47**
8 Spelling								1	0.35**	−0.49**	−0.66**	0.39**	0.58**
9 Text writing rate copying									1	−0.21*	−0.26**	0.20*	0.22*
10 Reading comprehension										1	0.56**	−0.80**	−0.50**
11 Written expression											1	−0.50**	−0.78**
12 Reading comprehension risk												1	0.49**
13 Written expression risk													1

Note: ILT—Invented Letter Task. Rate and accuracy. **, correlation is significant at the .001 level (2-tailed). *, significant at the .01 level (2-tailed).

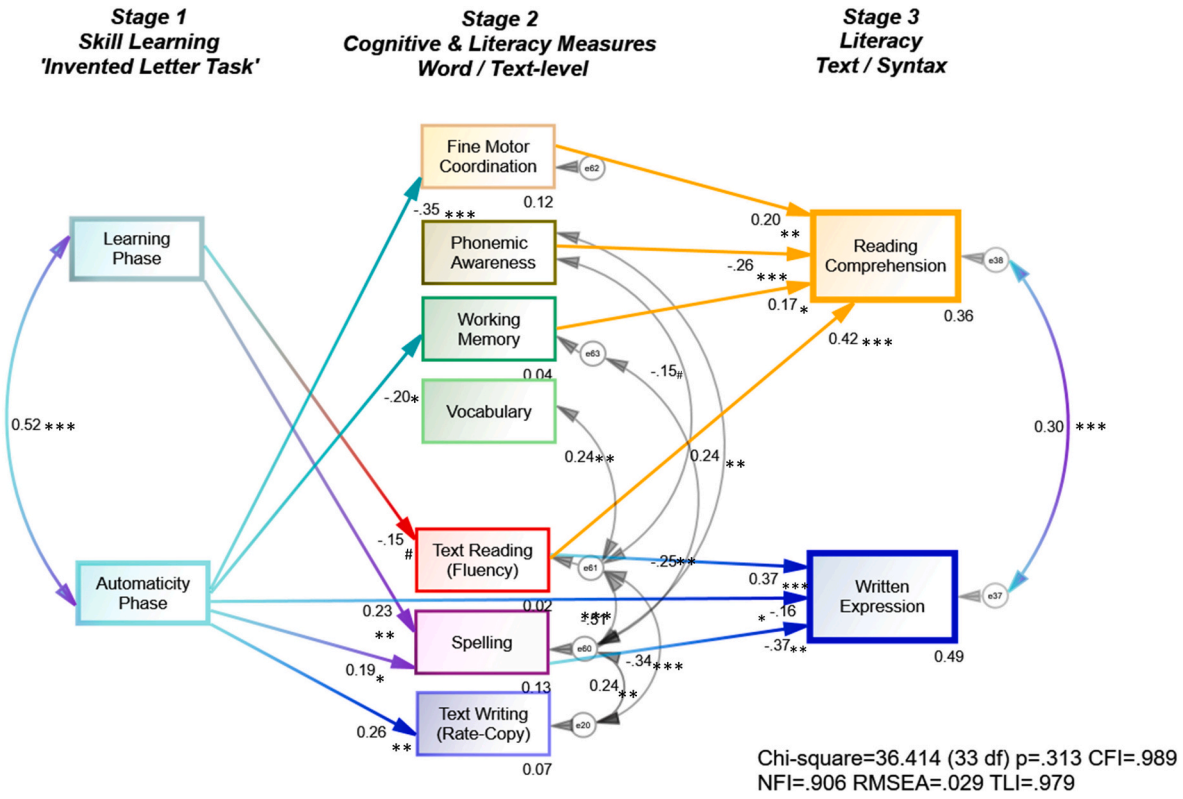


Fig. 5. Autoregressive SEM for reading comprehension and written expression. Depicting longitudinal relations among Stage 1, Skill learning Invented Letter Task (ILT) (learning and automaticity phases); Stage 2, Cognitive and Literacy Measures; and Stage 3, Literacy Text/Syntax. Straight-coloured arrows represent significant relations with the predicted variable coloured as is the arrow. The β is written under each arrow. Correlations are shown on the curved arrows (Pearson's coefficients-r). $^{\#}p = .08$, $*p < .05$, $**p < .01$, $***p < .001$.

significance as Fig. 5 with one main difference – working memory did not predict reading comprehension risk. In Fig. 6, the final data fit measures are: $\chi^2(28)=28.23$, $p=.45$, $N=124$; CFI=0.99; RMSEA=0.01; TLI=0.99; NFI=0.89. Table A3 in the Appendix includes standardized estimates of regression coefficients with their p values, the latter also indicated in the figure. The R^2 values indicate that the model explained 29.0% of the variance in reading-comprehension risk ($p < .05$).

In Fig. 7, the final data fit measures are: $\chi^2(28)=23.54$, $p=.70$, $N=124$; CFI=1.00; RMSEA=0.00; TLI=1.014; NFI=0.91. Table A3 in the Appendix includes standardized estimates of regression coefficients with their p values, the latter also indicated in the figure. The R^2 values indicate that the model explained 36.0% of the variance in written expression ($p < .05$). The same direct and indirect paths appeared for written expression (see Fig. 5).

4. Discussion

Practice plays a crucial role in literacy acquisition, yet its interaction with underlying linguistic, cognitive, and motor abilities has not been fully explored. While previous studies have examined the role of these underlying abilities in the acquisition of reading and writing, as well as risk for impairments in these abilities, the influence of skill learning in conjunction with these underlying abilities remains largely unstudied, particularly in children from low-SES backgrounds.

To fill this knowledge gap, the current study combined theoretical frameworks to create a measurable model. The proposed theoretical model (Fig. 1) is grounded in conceptual frameworks of literacy acquisition: Ullman's (2004) and Ullman et al.'s (2020) PDH model, associating the PLS with reading, Nicolson's and Fawcett's (2019) DNC

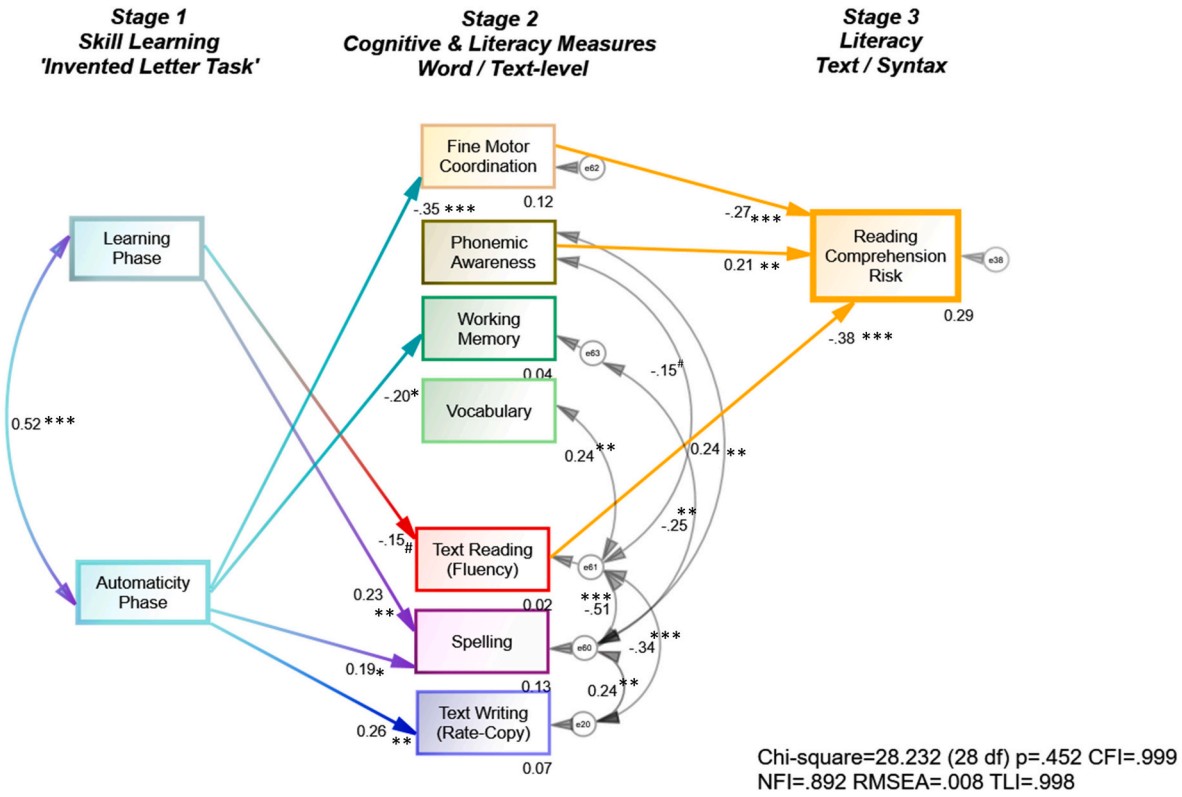


Fig. 6. Autoregressive SEM risk for impairment in reading comprehension Depicting longitudinal relations among Stage 1, Skill learning Invented Letter Task (ILT) (learning and automaticity phases); Stage 2, Cognitive and Literacy Measures; and Stage 3, Literacy Text/Syntax. Straight-coloured arrows represent significant relations with the predicted variable coloured as is the arrow. The β is written under each arrow. Correlations are shown on the curved arrows (Pearson's coefficients-r). $^{\#}p = .08$, $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

framework suggesting deficits in skill automatization as a basis for learning disorders, and Berninger, Swanson, and Griffin's (2015) cascading levels of language framework emphasize the levels of subword, word, and text-syntax for mastering reading comprehension and written expression. The current study integrates these theories of literacy acquisition based on preliminary work by Julius et al. (2016), which showed longitudinal correlations among the ILT, reading, and writing in monolingual Hebrew-speaking kindergarten to second-grade students from middle-to-high SES.

The proposed model tested the learning and automaticity phases of a novel letter-writing task (the ILT) and the direct and indirect prediction of reading comprehension and written expression, as well as risk for impairments in these abilities, in conjunction with known contributors, including phonological awareness, vocabulary, fine motor skills, working memory, and literacy abilities. The model involves multiple levels of analysis (i.e., letter, word, and text/syntax). In this model, the ILT learning phases represent a metric of procedural learning and a proxy for the subword (letter writing) level.

The notion that the procedural learning system may contribute to language learning is not new (Reber, 1967; Tomblin et al., 2007). However, can it contribute to higher levels of language, such as syntax? The findings confirmed the model and validated the hypothesis that the ILT learning and automaticity phases directly and indirectly predict a variety of literacy skills, including reading fluency, writing rate, spelling, written expression, and the risk for impairment in written expression (Table 3, Fig. 5). The model presented different paths involving specific cognitive and literacy measures (Kim et al., 2011; Lê et al., 2021).

The study focused on children from low SES backgrounds. These children face many challenges that can impact their learning, such as a lack of stimulating and safe environments and limited access to educational resources required for developing life skills such as language, fine motor, and literacy skills. Low SES is strongly associated with deficits in the intentional acquisition of knowledge through declarative memory (Farah, 2017), but not with the acquisition of practice-dependent skills (Hollander & Adi-Japha, 2021; Leonard et al., 2015). Therefore, the finding of low literacy skills in these children does not stem from impaired learning following practice, but rather from a lesser amount of required practice due to the lack of a stimulating environment. The current study stresses the involvement of the PLS in the acquisition of complex written language skills, suggesting that ample practice of PLS-related underlying skills (e.g., fine motor skills and reading fluency) in the schooling system may partially compensate for a home environment that possibly does not provide such practice (Kim et al., 2023; Rodriguez-Oramas et al., 2022(. Guidance to parents on the importance and effectiveness of timely practice may be beneficial, as the practice of basic skills (e.g., fine motor skills) should precede that of more complex skills (e.g., handwriting, Ghanamah et al., 2024). It is important to note, however, that this study's findings are based on a specific sample of low-SES second graders, and their generalizability may vary depending on cultural, linguistic, or socioeconomic contexts.

4.1. Differences between learning and automaticity phases in skill learning

Effective learning of a new skill involves an intense training phase that is followed by consolidation processes in which the gains accrued are stabilized against interference and retained for weeks, months, or even years. The learning phase for a simple task may reflect an intermediate asymptotic level of performance, whereas the automaticity phase may reflect the consolidation of the learnt skill in the long-term memory system. The differences in performance levels between the end of the training phase and the automaticity phase reflect the attainment of abilities (Dudai, 2004; Dayan & Cohen, 2011).

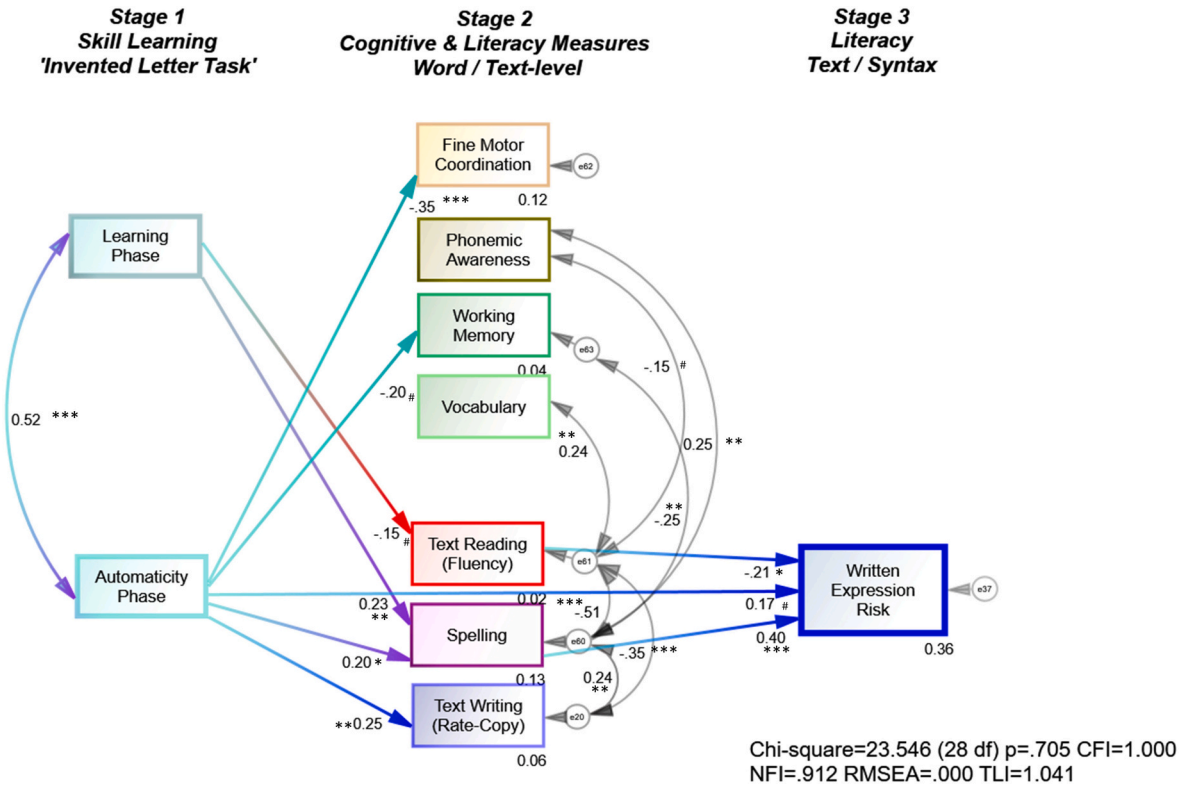


Fig. 7. Autoregressive SEM risk for impairment in written expression. Depicting longitudinal relations among Stage 1, Skill learning Invented Letter Task (ILT) (learning and automaticity phases); Stage 2, Cognitive and Literacy Measures; and Stage 3, Literacy Text/Syntax. Straight-coloured arrows represent significant relations with the predicted variable coloured as is the arrow. The β is written under each arrow. Correlations are shown on the curved arrows (Pearson's coefficients-r). $^{\#}p = .08$, $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

4.2. Skill-learning and literacy acquisition

Ullman (2004), Ullman and Pullman (2015) and Nicolson and Fawcett (2007, 2019) highlighted the role of the PLS as an important contributor to effective underlying processes of reading and writing acquisition (e.g., phonology and syntax), stressing the association between impaired PLS and impairment in reading and writing acquisition. Berninger et al. (2012), Berninger, Richards, and Abbott (2015), and Berninger, Swanson, and Griffin (2015) suggest a model with a typology of language levels to explain children's abilities to acquire higher written-language (i.e., literacy) skills at the syntax and text levels. These higher levels of written language are based on lower-level literacy attainments at the subword and word levels (i.e., 'cascading levels of language'). Although Berninger and colleagues include measures of letter writing automaticity in predicting reading and writing outcomes (e.g., Berninger et al., 2002), the PLS was not included in the conceptual framework. Therefore, the current study extends this model by including PLS measures reflecting the two phases: learning and automaticity. Conversely, Ullman's (2004) & Ullman et al.'s (2020) theory of the role of the PLS in reading acquisition, and Nicolson and Fawcett's (2019) DNC framework for dyslexia do not involve domain-specific measures of learning abilities. In the current study, these theories are integrated, and the ILT plays a specific role: (1) as a domain-general PLS predictor (representing both "learning" and "automaticity"), and (2) as a domain-specific predictor of literacy skills focused at the subword (letter) level, representing letter-writing acquisition.

4.3. Model findings

4.3.1. The ILT learning phase

The ILT learning phase reflects the fast learning phase of a novel skill, following initial but substantive practice, until asymptotic performance is reached. Further improvement is expected with additional practice.

The ILT learning phase was correlated with written expression and the risk for impairment in written expression, but not with reading comprehension or the risk for impairment in reading comprehension. The SEM model findings indicate that the ILT learning phase was only indirectly associated with reading comprehension through text reading fluency (see also Fuchs et al., 2001; Kim et al., 2010; Valencia et al., 2010 for the role of text reading fluency in reading comprehension). Furthermore, the ILT learning phase only indirectly predicted written expression (as well as risk for impairment in this ability) through both text reading fluency and spelling (see also Berninger et al., 2002 for similar contributions of reading fluency and spelling to written expression).

4.3.2. The ILT automaticity phase

Automaticity is an effective skill-acquisition phase (less effort, less explicit need for attentional resources, higher rate, and higher accuracy). Nicolson and Fawcett (1990), focusing on reading, proposed within the framework of automatization deficit that dyslexic children have difficulties in making any skill automatic, be it a cognitive skill such as reading or a motor skill such as catching a ball. Nicolson et al. (2001) proposed a development causal chain, in which a range of factors may be at play in the pre-reading years, for example, associating motor skill learning impairment with writing impairments (Nicolson & Fawcett, 2011).

The SEM model findings suggest that the ILT automaticity phase was associated with reading comprehension only indirectly through fine motor skills and working memory. Previous studies indicate the contribution of fine motor skills and working memory to reading skills or reading impairments (Costa et al., 2018; Dinehart & Manfra, 2013; Grissmer et al., 2010). The SEM model further indicated that the ILT automaticity phase predicted the risk for reading comprehension only

indirectly through fine motor skills. Note that although working memory predicted reading comprehension, it did not predict the risk for impairment in reading comprehension, supporting the results of previous studies (Schiff et al., 2016; Shalev-Leifer, 2016), thereby stressing the linguistic origin of reading impairments at the syntax level.

Interestingly, the ILT automaticity phase was correlated with the risk for reading comprehension, but *not* with reading comprehension. It may be suggested that the correlation with the risk measure emerged due to the crucial role of practice in attaining reading comprehension, and once enough initial practice was achieved, other factors (notably cognitive and linguistic) explain reading comprehension. Alternatively, the correlation with the risk measure may have emerged due to common underlying surface components as the horizontal move between discrete elements required in the ILT and in reading.

The ILT automaticity phase was correlated with written expression and the risk for impairment in written expression. As expected (Graham et al., 1997), In the SEM model, ILT automaticity directly predicted written expression (as well as risk for impairment in this ability) and contributed indirectly to both the skill and the risk for impairment in this skill through spelling (Berninger et al., 2002, Niedo, Abbott, and Berninger, 2014).

4.3.3. Additional predictors of reading comprehension and written expression

Beyond the contribution of fine motor skills, working memory, and text reading fluency to reading comprehension mentioned above, an additional contribution to reading comprehension came from phonemic awareness (Ramus & Szenkovits, 2008; Stanovich, 1988). Phonemic awareness requires attentional control and is less strongly associated with skilled ILT performance. Regarding written expression, all contributions were related to ILT phases, either directly (letter writing automaticity, Graham et al., 1997) or indirectly through text reading fluency and spelling (Berninger et al., 2002). Thus, overall, the prediction of reading compression and written expression (as well as risk for impairment in these abilities) aligns with that of previous studies.

4.3.4. Concomitant prediction of reading comprehension and written expression

Children learn to read and write simultaneously, and these language systems mutually develop, supported by verbal expressive and comprehension abilities. As in previous studies, the current investigation sought to predict these co-developing systems as two dependent outcomes. The analysis indicated that single-system associations were maintained, and an additional association between the two systems emerged. These findings suggest that contributions to each system (including ILT contributions) are robust and that although reading and writing draw on shared knowledge, they are separable skills (Berninger et al., 2002; Fitzgerald & Shanahan, 2000). The SEM of the two predicted abilities indicated a larger explained variance than the sum of the two separate models. Although text reading fluency contributed to both abilities, the text writing rate did not contribute directly to any of these abilities, possibly due to the involvement of the ILT. These findings emphasize the importance of early reading to writing skills, above letter automaticity (Julius et al., 2016).

4.3.5. ILT associations with literacy skills

As in previous studies (Ghanamah et al., 2024; Julius et al., 2016; Salameh-Matar et al., 2024), ILT performance correlated with a variety of literacy skills. In addition to the copying rate, which was correlated with the learning and automaticity phases, the ILT was correlated with spelling (learning and automaticity phases), reading fluency (automaticity phase), written expression, and risk for this ability (learning and automaticity phases), as well as with risk for reading comprehension (automaticity phase). The ILT did not correlate with other linguistic and literacy skills, specifically, phonology, vocabulary, and reading comprehension. This stresses that the ILT is more strongly associated with writing and lower-level reading skills than with higher-order reading skills, as reading comprehension. Reading comprehension relies on additional factors, including vocabulary and prior knowledge, both of which are supported by declarative memory.

Previous studies did not always find a correlation between implicit/procedural memory measures and literacy measures in populations with a variety of abilities (e.g., West et al., 2018). As different studies use different literacy and memory measures, it may be suggested that procedural learning does not contribute equally to all literacy measures and that perhaps domain-specific PLS measures show higher correlations.

4.4. Limitations and future research

The conclusions of this study must be considered within their limitations. Literacy acquisition is a dynamic and multifaceted process shaped by a diverse range of cognitive, social, and environmental factors. An attempt was made to track some of the influential cognitive predictors of reading and writing across second graders. Children's abilities were not tracked further. Individual profile assessments to diagnose the indicators of their language level in higher grades, as Berninger, Swanson, and Griffin (2015) suggested, may advise more comprehensive model predictions and better explain their abilities.

The current study used a specific skill-learning task. There is a further need to determine if other types of procedural learning, such as verbal learning and statistical learning, behave differently than motor learning in such a research design. For the practical conclusions of this study, it is important to recall the two different but complementary views of the ILT that were used: as a PLS representative; and as a subword (letter) domain-specific proxy for higher language levels. Other studies suggest that domain-specific learning tasks may better differentiate learning in DD than other tasks (Gabay et al., 2012). Thus, future studies should further compare predictions made by procedural learning tasks with higher ecological validity vs. more typical PLS tasks as the serial reaction time task.

It is important to note that this study investigated the risk for impairments in reading comprehension and written expression in Hebrew-speaking children. One of the unique characteristics of Hebrew is that it is a highly syntactic language, exhibiting a rich and dense morphology with a word structure that expresses a rich array of grammatical and semantic notions (Primor et al., 2011; Ravid, 2012). Future studies should compare the use of ILT among children who speak other languages (Ghanamah et al., 2024), seeking differences in the effect of procedural learning predictors on the risk for impairment in reading comprehension at young ages.

The findings of associations between early literacy abilities at different levels are consistent with findings of Berninger, Swanson, and Griffin (2015) among higher-SES populations, suggesting that the main tendencies of abilities among children from low-SES populations resemble those of higher-SES youngsters. Yet, further research should specify issues such as maternal education that may impact children's ILT performance and academic achievements, thereby serving as an underlying factor explaining the association between these variables (Hollander & Adi-Japha, 2021).

5. Conclusions and classroom implications

The current study validated a theoretical model associating domain-general PLS phases of learning and automaticity with literacy skills, using a domain-specific measure, the ILT. The SEM model showed that performance on the ILT, directly and indirectly, was associated with reading and writing measures in low-SES second-grade students. A previous study in middle-to-high SES population showed associations of ILT phases to reading and writing in kindergarten-to-second-grade students, stressing the importance of skill practice to reading and writing *in all SES levels*. Nevertheless, these findings should be interpreted with caution, as results may not generalize across languages or orthographies,

and future research should examine similar models in more diverse settings.

The study implicates the importance of basic procedural skills in developing higher-level literacy abilities. For example, the study stresses the importance of well-practiced letter-writing as assessed using the ILT. This result resonates with existing literature. The practice of basic letter-writing skills, as repeated practice of symbol and letter writing, may improve writing fluency. For example, practice of 6 blocks of the ILT for four consecutive days improved handwriting fluency in second graders up to 4–5 weeks post-training, where the writing test was preceded by 2 ILT blocks (Ghanamah & Adi-Japha, 2023). Symbol and letter writing training were shown to improve letter formation and writing automaticity, the latter transferring to improved word reading in teacher-referred first graders (Berninger et al., 2006).

Furthermore, learning and automaticity of the ILT had different contributions to reading and writing, depending on the level of the acquired literacy skill. Skill levels may improve with practice, but partial learning of underlying skills may impede the development of higher-level skills. For example, lower reading fluency may impede reading comprehension as well as written expression.

Second grade is a dynamic period of emerging and rapid change in children's literacy-skill proficiency as extensive practice takes place and affects skill acquisition. At this age, children tackle higher levels of literacy challenges but are still in the process of learning. This leaves doubt as to whether children who face difficulties have a specific learning disorder or are typical learners presenting a delay. Nevertheless, early detection and intervention are warranted (Berninger et al., 2006). Children in kindergarten may be tested using a skill-learning screening task (such as the ILT). We suggest using the ILT for identifying children at risk for "practice-related literacy learning." Such testing may serve as a concurrent progress monitor and also predict slower learning of later literacy skills, calling for early intervention. Specifically, initial ILT (2–4 blocks) speed and accuracy correlated with the concurrent and next year handwriting performance (speed and accuracy) of Arabic (Ghanamah et al., 2024; Salameh-Matar et al., 2024) and Hebrew (Julius et al., 2016) speaking kindergarten and second-grade children of a varied SES background. Furthermore, Julius et al. (2016) showed that ILT performance speed measured 24 h after practice correlated with both contemporaneous handwriting speed and handwriting and reading speed in the subsequent year, beyond the effects of age, socioeconomic status, and initial ILT performance speed, highlighting the impact of assessing *learning* for a better prediction of achievements. Overall, the findings of the current study stress the importance of the PLS for skill acquisition and implicate ILT assessment for identifying early risk and monitoring progress of basic literacy skills at the letter level.

CRediT authorship contribution statement

Chagit Hollander: Writing – original draft, Software, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization, Writing – review & editing. **Shoshi Dorfberger:** Validation, Software. **Michal Hochhauser:** Visualization. **Esther Adi-Japha:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis.

Ethics statement

The studies involving human participants were reviewed and approved by the Israel Ministry of Education Chief Scientist Institutional Review Board of Ethics (MoE) (7883/054 10.32), following human subject's guidelines/principles for all research held in the Israeli educational system. Parents signed the Ministry of Education consent forms.

Declaration of Generative AI and AI-assisted technologies in the writing process

We confirm that we have taken note of the information on using generative AI in scientific writing. We confirm that we have disclosed any use of generative AI directly in the manuscript.

Declaration of competing interest

The authors have no competing interests to report.

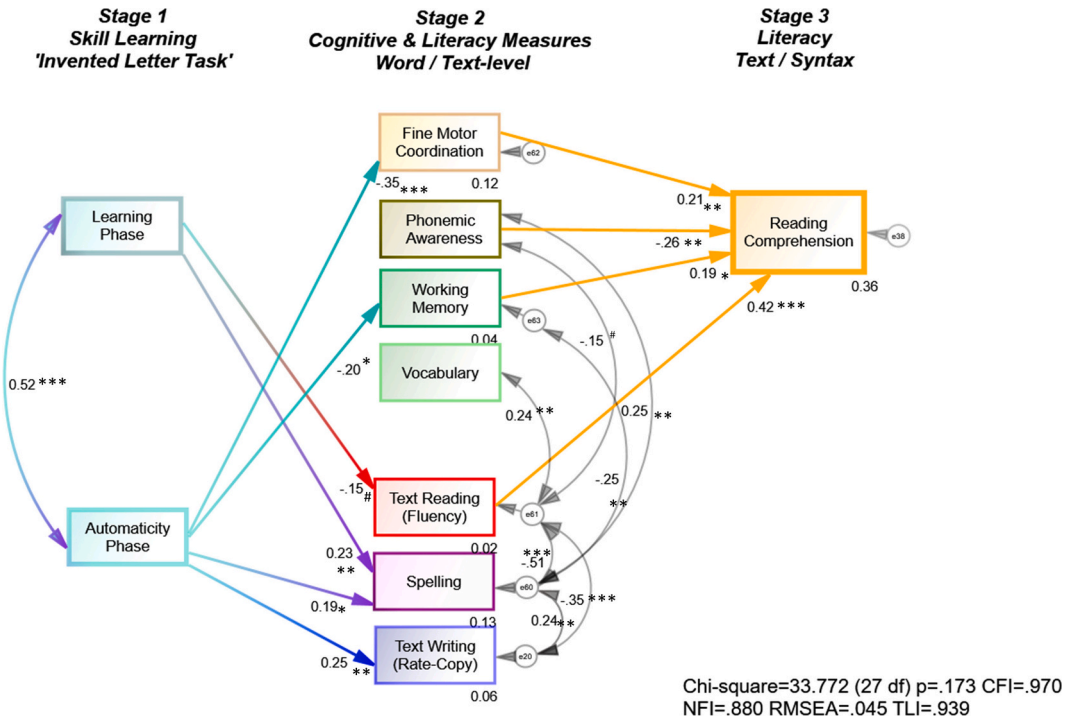
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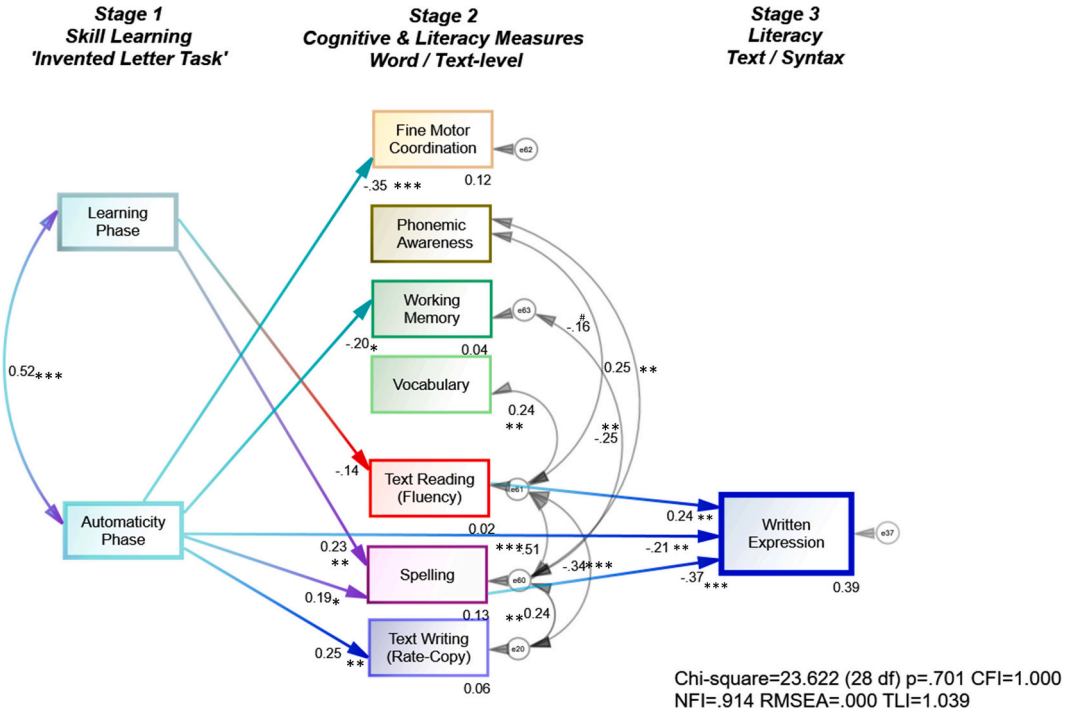
Appendix 1

Table 1A
Pearson correlations of all variables in the study with age and Z-SES.

Correlations		
	Pearson correlation	
	Age in months	Mean Z SES
1. Learning phase (ILT)	0.03	0.06
2. Automaticity phase (ILT)	−0.04	0.18
3. Fine motor coordination	0.17	−0.17
4. Phonemic awareness	0.20*	−0.06
5. Working memory	−0.11	0.03
6. Vocabulary	0.04	−0.05
7. Text reading (fluency)	−0.10	0.04
8. Spelling	0.04	0.04
9. Text writing rate (copying task)	0.06	0.05
10. Reading comprehension	−0.04	0.03
11. Written expression	−0.08	0.02
12. Risk for impairments in reading comprehension	0.02	0.06
13. Risk for impairments written expression	−0.04	0.03



Appendix Fig. 1A. Autoregressive SEM separate model for reading comprehension. Depicting longitudinal relations among Stage 1, Skill learning Invented Letter Task (ILT) (learning and automaticity phases); Stage 2, Cognitive and Literacy Measures; and Stage 3, Literacy Text/Syntax. Straight-coloured arrows represent significant relations with the predicted variable coloured as is the arrow. The β is written under each arrow. Correlations are shown on the curved arrows (Pearson's coefficients- r). $^{\#}p = .08$, $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.



Appendix Fig. 1B. Autoregressive SEM separate model for written expression. Depicting longitudinal relations among Stage 1, Skill learning Invented Letter Task (ILT) (learning and automaticity phases); Stage 2, Cognitive and Literacy Measures; and Stage 3, Literacy Text/Syntax. Straight-coloured arrows represent significant relations with the predicted variable coloured as is the arrow. The β is written under each arrow. Correlations are shown on the curved arrows (Pearson's coefficients- r). $^{\#}p = .08$, $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

Appendix 2. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.lindif.2025.102779.

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